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Lab 1: Network Commands & IP Addressing

Checking Physical (MAC) Address

1. What command shows all network details including the physical (MAC) address?
 - `ipconfig /all`

```
Default Gateway . . . . . : 192.168.1.1

C:\Users\Administrator>ipconfig /all

Windows IP Configuration

Host Name . . . . . : WIN-BGD06Q5RE0H
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : 3C-2C-30-F4-EF-E3
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : D8-F2-CA-72-17-36
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : DA-F2-CA-72-17-35
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wireless-AC 9462
Physical Address. . . . . : D8-F2-CA-72-17-35
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::be03:2d4f:fd65:d39e%18(Preferred)
```

```

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Wireless-AC 9462
Physical Address. . . . . : D8-F2-CA-72-17-35
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . : fe80::be03:2d4f:fd65:d39e%18(Preferred)
IPv4 Address. . . . . : 192.168.1.79(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Monday, November 17, 2025 10:22:59 AM
Lease Expires . . . . . : Tuesday, November 18, 2025 11:01:37 AM
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAD . . . . . : 165212874
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-9C-61-BA-3C-2C-30-F4-EF-E3
DNS Servers . . . . . : fe80::8b8:d608:339c:3c27%18
                        192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Administrator>

```

2. How can you see your device's MAC address?

- Getmac

```

C:\Users\Administrator>getmac

Physical Address      Transport Name
=====
3C-2C-30-F4-EF-E3    Disconnected
D8-F2-CA-72-17-35    \Device\Tcpip_{F3AA62B1-B7AF-4FB6-BC11-F6FA5DCC3EEC}

C:\Users\Administrator>getmac /v

Connection Name Network Adapter Physical Address      Transport Name
=====
Ethernet        Realtek PCIe Gb 3C-2C-30-F4-EF-E3    Disconnected
Wi-Fi           Intel(R) Wirele D8-F2-CA-72-17-35    \Device\Tcpip_{F3AA62B1-B7AF-4FB6-BC11-F6FA5DCC3EEC}

C:\Users\Administrator>_

```

3. How can you show the MAC address with more details?

- getmac /v

```

C:\Users\Administrator>getmac

Physical Address      Transport Name
=====
3C-2C-30-F4-EF-E3    Disconnected
D8-F2-CA-72-17-35    \Device\Tcpip_{F3AA62B1-B7AF-4FB6-BC11-F6FA5DCC3EEC}

C:\Users\Administrator>getmac /v

Connection Name Network Adapter Physical Address      Transport Name
=====
Ethernet        Realtek PCIe Gb 3C-2C-30-F4-EF-E3    Disconnected
Wi-Fi           Intel(R) Wirele D8-F2-CA-72-17-35    \Device\Tcpip_{F3AA62B1-B7AF-4FB6-BC11-F6FA5DCC3EEC}

C:\Users\Administrator>_

```

APIPA & IP Configuration

4. What is the default APIPA address when the device fails to get an IP from DHCP?

- Starts with: 169.254.x.x
device success to get dhcp ip so apiipa not exist

```
C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . : fe80::be03:2d4f:fd65:d39e%18
    IPv4 Address. . . . . : 192.168.1.79
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\Administrator>ipconfig
```

5. What command shows your current IP settings?

- o Ipconfig

```
Microsoft Windows [Version 10.0.22631.5909]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . . : fe80::be03:2d4f:fd65:d39e%18
    IPv4 Address. . . . . : 192.168.1.79
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\Administrator>
```

6. How do you release your current IP address?

- ipconfig /release

```
C:\Users\Administrator>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::be03:2d4f:fd65:d39e%18
    Default Gateway . . . . . :

C:\Users\Administrator>
```

7. How do you request a new IP address from DHCP?

- ipconfig /renew

```
C:\Users\Administrator>ipconfig /renew

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconnected.
No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::be03:2d4f:fd65:d39e%18
    IPv4 Address. . . . . : 192.168.1.79
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.1.1

C:\Users\Administrator>
```

Ping Commands

9. How do you test connection to an IP address?

- o ping [IP address]

```
C:\Users\Administrator>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=44ms TTL=117
Reply from 8.8.8.8: bytes=32 time=44ms TTL=117
Reply from 8.8.8.8: bytes=32 time=44ms TTL=117
Reply from 8.8.8.8: bytes=32 time=44ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 44ms, Maximum = 44ms, Average = 44ms

C:\Users\Administrator>ping 8.8.8.8 -l 1000

Pinging 8.8.8.8 with 1000 bytes of data:
Reply from 8.8.8.8: bytes=1000 time=156ms TTL=117
Reply from 8.8.8.8: bytes=1000 time=57ms TTL=117
Reply from 8.8.8.8: bytes=1000 time=128ms TTL=117
Reply from 8.8.8.8: bytes=1000 time=59ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 57ms, Maximum = 156ms, Average = 100ms

C:\Users\Administrator>
```

10. How do you test connection to a website (URL)?

- ping www.example.com

```
C:\Users\Administrator>ping www.google.com -n 10

Pinging www.google.com [142.250.201.4] with 32 bytes of data:
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=59ms TTL=117
Reply from 142.250.201.4: bytes=32 time=46ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=47ms TTL=117
Reply from 142.250.201.4: bytes=32 time=53ms TTL=117

Ping statistics for 142.250.201.4:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 59ms, Average = 47ms

C:\Users\Administrator>ping www.google.com -t

Pinging www.google.com [142.250.201.4] with 32 bytes of data:
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=47ms TTL=117
Reply from 142.250.201.4: bytes=32 time=86ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=182ms TTL=117
Reply from 142.250.201.4: bytes=32 time=319ms TTL=117
Reply from 142.250.201.4: bytes=32 time=320ms TTL=117
Reply from 142.250.201.4: bytes=32 time=139ms TTL=117

Ping statistics for 142.250.201.4:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 320ms, Average = 127ms
Control-C
^C
C:\Users\Administrator>
```

11. How do you send a ping with a specific data size?

- ping [IP]-l [size]

```
C:\Users\Administrator>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=44ms TTL=117
Reply from 8.8.8.8: bytes=32 time=44ms TTL=117
Reply from 8.8.8.8: bytes=32 time=44ms TTL=117
Reply from 8.8.8.8: bytes=32 time=44ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 44ms, Average = 44ms

C:\Users\Administrator>ping 8.8.8.8 -l 1000

Pinging 8.8.8.8 with 1000 bytes of data:
Reply from 8.8.8.8: bytes=1000 time=156ms TTL=117
Reply from 8.8.8.8: bytes=1000 time=57ms TTL=117
Reply from 8.8.8.8: bytes=1000 time=128ms TTL=117
Reply from 8.8.8.8: bytes=1000 time=59ms TTL=117

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 57ms, Maximum = 156ms, Average = 100ms

C:\Users\Administrator>
```

12. How do you send a specific number of ping requests?

- ping [IP]-n [count]

```
C:\Users\Administrator>ping www.google.com -n 10

Pinging www.google.com [142.250.201.4] with 32 bytes of data:
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=59ms TTL=117
Reply from 142.250.201.4: bytes=32 time=46ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=47ms TTL=117
Reply from 142.250.201.4: bytes=32 time=53ms TTL=117

Ping statistics for 142.250.201.4:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 59ms, Average = 47ms

C:\Users\Administrator>ping www.google.com -t

Pinging www.google.com [142.250.201.4] with 32 bytes of data:
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=47ms TTL=117
Reply from 142.250.201.4: bytes=32 time=86ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=182ms TTL=117
Reply from 142.250.201.4: bytes=32 time=319ms TTL=117
Reply from 142.250.201.4: bytes=32 time=320ms TTL=117
Reply from 142.250.201.4: bytes=32 time=139ms TTL=117

Ping statistics for 142.250.201.4:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 320ms, Average = 127ms
Control-C
^C
C:\Users\Administrator>
```

13. How do you send continuous ping requests until you stop it manually?

- ping [IP]-t

```
C:\Users\Administrator>ping www.google.com -n 10

Pinging www.google.com [142.250.201.4] with 32 bytes of data:
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=59ms TTL=117
Reply from 142.250.201.4: bytes=32 time=46ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=47ms TTL=117
Reply from 142.250.201.4: bytes=32 time=53ms TTL=117

Ping statistics for 142.250.201.4:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 59ms, Average = 47ms

C:\Users\Administrator>ping www.google.com -t

Pinging www.google.com [142.250.201.4] with 32 bytes of data:
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=44ms TTL=117
Reply from 142.250.201.4: bytes=32 time=47ms TTL=117
Reply from 142.250.201.4: bytes=32 time=86ms TTL=117
Reply from 142.250.201.4: bytes=32 time=45ms TTL=117
Reply from 142.250.201.4: bytes=32 time=182ms TTL=117
Reply from 142.250.201.4: bytes=32 time=319ms TTL=117
Reply from 142.250.201.4: bytes=32 time=320ms TTL=117
Reply from 142.250.201.4: bytes=32 time=139ms TTL=117

Ping statistics for 142.250.201.4:
    Packets: Sent = 10, Received = 10, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 44ms, Maximum = 320ms, Average = 127ms
Control-C
^C
C:\Users\Administrator>
```

Netstat

19. How do you show all active connections?

- o netstat-a

```
C:\Users\Administrator>netstat -a
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:445	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:3306	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:7680	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:33060	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:49664	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:49665	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:49666	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:49667	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:49669	WIN-BGD06Q5REOH:0	LISTENING
TCP	0.0.0.0:63452	WIN-BGD06Q5REOH:0	LISTENING
TCP	127.0.0.1:1001	WIN-BGD06Q5REOH:0	LISTENING
TCP	127.0.0.1:49564	WIN-BGD06Q5REOH:49674	ESTABLISHED
TCP	127.0.0.1:49674	WIN-BGD06Q5REOH:0	LISTENING
TCP	127.0.0.1:49674	WIN-BGD06Q5REOH:49564	ESTABLISHED
TCP	127.0.0.1:49679	WIN-BGD06Q5REOH:49680	ESTABLISHED
TCP	127.0.0.1:49680	WIN-BGD06Q5REOH:49679	ESTABLISHED
TCP	127.0.0.1:49681	WIN-BGD06Q5REOH:49682	ESTABLISHED
TCP	127.0.0.1:49682	WIN-BGD06Q5REOH:49681	ESTABLISHED
TCP	192.168.1.79:139	WIN-BGD06Q5REOH:0	LISTENING
TCP	192.168.1.79:49219	ec2-35-162-42-157:https	ESTABLISHED
TCP	192.168.1.79:49284	mad01s25-in-f195:https	ESTABLISHED
TCP	192.168.1.79:49303	146.75.53.32:https	ESTABLISHED
TCP	192.168.1.79:49340	ec2-44-240-219-187:https	ESTABLISHED
TCP	192.168.1.79:50180	mrs08s03-in-f22:https	ESTABLISHED
TCP	192.168.1.79:50252	20.150.89.196:https	ESTABLISHED
TCP	192.168.1.79:50253	20.42.73.28:https	ESTABLISHED
TCP	192.168.1.79:50254	52.110.15.195:https	TIME_WAIT
TCP	192.168.1.79:50255	52.182.143.215:https	ESTABLISHED
TCP	192.168.1.79:50256	52.168.117.170:https	ESTABLISHED
TCP	192.168.1.79:50360	server-18-161-111-52:https	TIME_WAIT

Shodan Maps Images Monitor Developer More...

SHODAN Explore Pricing 52.182.143.215

52.182.143.215 Regular View Raw Data Timeline Whois

// TAGS: cloud

General Information

Hostnames: events.data.microsoft.com, mobile.events.data.microsoft.com, vortex.data.microsoft.com, events.data.msn.cn, events.data.msn.com, pipe.skype.com

Domains: microsoft.com, msn.cn, msn.com, skype.com

Cloud Provider: Azure

Cloud Service: OneDsCollector

Country: United States

Open Ports

443

// 443 / TCP

HTTP/1.1 404 Not Found
Content-Length: 0
Server: Microsoft-HTTPAPI/2.0
Strict-Transport-Security: max-age=31536000
Date: Fri, 07 Nov 2025 20:37:42 GMT

SSL Certificate

Certificate:
Data:
Version: 3 (Rev2)

20. How do you show active connections with numbers only?

- o netstat-n

```
C:\Users\Administrator>netstat -n
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49679	127.0.0.1:49680	ESTABLISHED
TCP	127.0.0.1:49680	127.0.0.1:49679	ESTABLISHED
TCP	127.0.0.1:49681	127.0.0.1:49682	ESTABLISHED
TCP	127.0.0.1:49682	127.0.0.1:49681	ESTABLISHED
TCP	192.168.1.79:49576	135.225.242.64:443	ESTABLISHED
TCP	192.168.1.79:50908	44.237.199.172:443	ESTABLISHED
TCP	192.168.1.79:51205	52.123.200.65:443	ESTABLISHED
TCP	192.168.1.79:51721	98.66.133.184:443	ESTABLISHED
TCP	192.168.1.79:55382	172.187.61.142:443	ESTABLISHED
TCP	192.168.1.79:55383	52.110.15.203:443	TIME_WAIT
TCP	192.168.1.79:55384	52.110.15.203:443	TIME_WAIT
TCP	192.168.1.79:57337	52.111.231.2:443	ESTABLISHED
TCP	192.168.1.79:60230	52.111.231.50:443	TIME_WAIT
TCP	192.168.1.79:60407	44.237.199.172:443	ESTABLISHED
TCP	192.168.1.79:62325	52.107.243.2:443	ESTABLISHED
TCP	192.168.1.79:62504	20.42.65.90:443	TIME_WAIT
TCP	192.168.1.79:62710	52.112.120.220:443	ESTABLISHED
TCP	192.168.1.79:64704	52.111.231.50:443	ESTABLISHED
TCP	192.168.1.79:65030	4.208.35.47:443	ESTABLISHED
TCP	192.168.1.79:65354	23.55.48.171:443	ESTABLISHED
TCP	192.168.1.79:65355	52.111.231.50:443	ESTABLISHED

```
C:\Users\Administrator>
```

21. What command shows MAC addresses of other devices on the network?

arp -a

```
C:\Users\Administrator> arp -a

Interface: 192.168.1.79 --- 0x12
Internet Address      Physical Address      Type
192.168.1.1           c0-fd-84-9c-3c-27     dynamic
192.168.1.200         58-7f-66-aa-5c-75     dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\Administrator>
```

22. What command deletes the ARP cache?

arp -d

```
C:\Users\Administrator> arp -a

Interface: 192.168.1.79 --- 0x12
Internet Address      Physical Address      Type
192.168.1.1           c0-fd-84-9c-3c-27     dynamic
192.168.1.200         58-7f-66-aa-5c-75     dynamic
192.168.1.255         ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Example:
C:\Users\Administrator> arp -d

C:\Users\Administrator>
```

23. To know which IP address belongs to a domain name:

nslookup Domain_Name

Example:

nslookup Yahoo.com

```
C:\Users\Administrator>nslookup Yahoo.com
Server:  csp3.zte.com.cn
Address:  fe80::8b8:d608:339c:3c27

Non-authoritative answer:
Name:    Yahoo.com
Addresses:  2001:4998:44:3507::8000
            2001:4998:124:1507::f001
            2001:4998:124:1507::f000
            2001:4998:24:120d::1:0
            2001:4998:44:3507::8001
            2001:4998:24:120d::1:1
            74.6.143.25
            74.6.231.21
            74.6.231.20
            74.6.143.26
            98.137.11.163
            98.137.11.164

C:\Users\Administrator>
```

24. To know which domain name belongs to a specific IP address:

Type this command:

nslookup IP_address

Example:

nslookup 87.248.113.14

24. To know which domain name belongs to a specific IP address, type this command:

```
C:\Users\Administrator>nslookup 87.248.113.14
```

```
Server: csp1.zte.com.cn
```

```
Address: fe80::8b8:d608:339c:3c27
```

```
Name: et23-1.bas1-1-edg.amb.yahoo.com
```

```
Address: 87.248.113.14
```

```
C:\Users\Administrator>
```

Example: